

Logos Experiment 14 Report

Large Scaling with Weight Tying + SwiGLU on OpenWebText 100k

Field	Value
Run name	logos_exp14_large_scaling_weight_tying_swiglu
Objective	Test whether the best Logos efficiency recipe scales to a 100M+ parameter model
Dataset	OpenWebText subset, 100,000 samples
Model family	Decoder-only GPT-style causal language model
Core recipe	Weight Tying + SwiGLU feed-forward network
Training budget	3000 iterations

1. Purpose of the Experiment

Experiment 14 was designed as the first large-scale validation run for the Logos project. Earlier experiments showed that Weight Tying + SwiGLU improves compact model quality and efficiency. Experiment 13 confirmed that this recipe scales to a 51.20M-parameter medium model. Experiment 14 pushes the same idea further to a 124.03M-parameter model trained on 100k OpenWebText samples.

The main scientific question is: does the best efficient recipe continue to reduce validation loss and perplexity when the model and dataset are scaled up?

2. Model Configuration

Component	Value
Architecture	Large decoder-only GPT-style Transformer
Parameters	124,031,232
Layers	12
Embedding dimension	768
Attention heads	12
Head dimension	64
Context length	512 tokens
Batch size	32
Tokens per step	16,384
Feed-forward network	SwiGLU
SwiGLU hidden size	2048
Normalization	LayerNorm
Weight tying	Enabled

3. Training Setup

Setting	Value
Dataset	OpenWebText subset
Number of samples	100,000
Tokenizer	GPT-2 tokenizer

Setting	Value
Objective	Next-token prediction / causal language modeling
Optimizer	AdamW
Learning rate	3e-4
Weight decay	0.1
Gradient clipping	1.0
Warmup iterations	150
Maximum iterations	3000
Evaluation interval	300

4. Final Results

Metric	Experiment 14 Result
Best iteration	3000
Best validation loss	4.9201
Final validation loss	4.9244
Final validation perplexity	137.60
Checkpoint size	473.20 MB
Peak VRAM	17.73 GB
Training time	745.83 sec (~12.43 min)
Generation speed	66.10 tokens/sec
Generation latency	15.13 ms/token
Average GPU power	289.28 W
Estimated GPU energy	59.76 Wh

5. Comparison with Experiment 13

Metric	Exp 13 Medium	Exp 14 Large	Direction
Parameters	51.20M	124.03M	Increased
Dataset samples	50k	100k	Increased
Context length	512	512	Same
Best validation loss	5.0266	4.9201	Improved
Final validation loss	5.0421	4.9244	Improved
Final perplexity	154.79	137.60	Improved
Checkpoint size	195.36 MB	473.20 MB	Larger
Peak VRAM	13.09 GB	17.73 GB	Higher but practical
Training time	6.35 min	12.43 min	Longer
Energy	29.70 Wh	59.76 Wh	Higher

6. Key Interpretation

- Experiment 14 successfully validates that the Weight Tying + SwiGLU recipe scales beyond the medium 51M-parameter setting.

- The large model achieved the best perplexity so far: 137.60, improving over Experiment 13 at 154.79.
- The model used only 17.73 GB peak VRAM, which is still far below the available 48 GB VRAM on the RTX 6000 Ada system.
- The quality improvement came with predictable costs: larger checkpoint, longer training time, and higher energy usage.
- The result strengthens the Logos paper direction because it shows that the discovered small-model efficiency recipe transfers to a larger parameter regime.

7. Scientific Conclusion

Experiment 14 is a successful large-scale validation of the Logos efficiency recipe. Scaling the Weight Tying + SwiGLU decoder-only model from 51.20M to 124.03M parameters and increasing the dataset from 50k to 100k OpenWebText samples reduced final validation loss from 5.0421 to 4.9244 and reduced perplexity from 154.79 to 137.60.

The experiment shows that the best Logos recipe is not only useful for compact models; it continues to provide better language-modeling capability when scaled. This supports the broader research goal of improving language model capability per parameter, memory byte, latency unit, checkpoint size, and watt.

8. Current Best Models After Experiment 14

Category	Best Model	Reason
Best small quality model	Experiment 11	Strong compact quality: PPL 202.82 with 19.24M params
Best small efficiency model	Experiment 12	PPL 210.00 with 18.54M params and lower cost
Best medium scaling model	Experiment 13	PPL 154.79 with 51.20M params
Best overall quality model so far	Experiment 14	Lowest PPL so far: 137.60 with 124.03M params

9. Recommended Next Step

The next decision should be based on available time. Since Experiment 14 still ended with the best validation loss at the final iteration, it may still be undertrained. A longer-training version of Experiment 14 would test whether the 124M model continues improving. If time is limited, create the updated master leaderboard and final paper structure now.

- Option A: Experiment 15 - longer training of Experiment 14 for 5000 iterations.
- Option B: Experiment 15 - larger-context test with context length 1024, using a reduced batch size.
- Option C: Stop experiments and write the final paper/report using Experiments 1-14.

10. Final One-Line Summary

Experiment 14 demonstrates that the Logos Weight Tying + SwiGLU recipe scales successfully to a 124M-parameter decoder-only language model, reaching the best validation perplexity so far while using only 17.73 GB of VRAM.